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Understanding the Emergence and Trajectory of Job Insecurity Due to Smart Technology, Artificial Intelligence, Robotics, and Automation

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ABSTRACT

Smart technologies, artificial intelligence, robotics, and automation (STARA) can revolutionize the labor market by substituting human labor. STARA Awareness has been introduced to capture employees' appraisal of the impact of STARA on their employment without a thorough validation and overarching theoretical framework. Therefore, we contributed a content validation of STARA Awareness and examined the internal structure of the suggested measurement instrument, the differentiation from cognitive and affective job insecurity (JI), potential antecedents, and its 1-year trend. We conducted two cross-sectional ($N_1 = 215$, $N_2 = 224$) and one longitudinal study ($N_3 = 233$) with German employees from various branches. We adapted the questionnaire and redefined the construct as affective automation-related job insecurity (AAJI) based on content criticism. Our results indicate that AAJI is weakly positively related to cognitive and affective JI but empirically different. We identified the substitution potential of occupation, the use of STARA as positive predictors, and core self-evaluations as a negative predictor of AAJI. Latent growth curve models reveal no linear change of AAJI over 1 year but different trajectories as a function of the use of STARA. Thus, AAJI represents a digitalization-specific form of job insecurity with its distinct nomological net and high temporal stability.

From autonomous self-service checkouts in retail, assistance systems in assembly, and collaborative robots in manufacturing to ChatGPT—smart technologies, artificial intelligence, robotics, and automation (STARA) (Brougham and Haar 2018; 2020) fundamentally revolutionize work processes and the labor market (Salmon et al. 2023; Walczok and Bipp 2024). Contrary to previous waves of industrialization, in which primary and secondary sectors were mainly affected by downsizing, the tertiary sector is increasingly downsized by changes in Industry 4.0 too (Brougham and Haar 2018). Evidence by Yam et al. (2023) indicates that innovative technologies, such as robots, evoke employees' *job insecurity* (JI) as an unintended negative effect of ongoing digitalization. To capture how employees appraise the impact of new technologies on their employment, Brougham and Haar (2018) introduced the construct

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of *STARA Awareness*. They defined it as "the extent to which an employee views the likelihood of Smart Technology, Artificial Intelligence, Robotics and Algorithms impacting on their future career prospects" (p. 3). While they provided initial empirical correlations of STARA Awareness with negative work-related outcomes (e.g., burnout, turnover intentions), thorough research on the nature of this construct, separation from the established JI, its antecedents, and its temporal stability is still scarce, despite the rising numbers of articles about the construct. Given that STARA Awareness encompasses cognitive and affective elements, this may result in misleading interpretations of current empirical evidence. In addition, STARA Awareness focuses on the substitution of whole jobs and does not account for the more realistic substitution of specific tasks within jobs due to the digital transformation of the labor market (Dengler and Matthes 2018). Hence, an in-depth investigation and refinement of this suggested construct is necessary to understand how technological advances impact employees.

Therefore, using three studies, we aim to further develop a content-valid digitalization-specific form of job insecurity based on the STARA Awareness construct and corresponding measurement instrument (Brougham and Haar 2018). We position STARA Awareness in the JI literature by revising its definition and measurement and by applying it to a comprehensive model of JI (Shoss 2017). Moreover, we aim to investigate its 1-year trend and identify potential antecedents for its intercept and its trajectory. In the Pilot Study and Study 1, we examine its internal structure on the measurement level, its separation from the established constructs of *cognitive JI* (perceived likelihood of losing one's job or attractive job features) and *affective JI* (worry and concern about losing one's job or attractive job features), and investigate potential antecedents. Therefore, we aim to transfer antecedents from JI research (indicators for technological change, individual differences) to STARA Awareness. For this, we provide an extension of the construct and questionnaire by also including the substitution of core tasks within jobs (Study 1). In Study 2, we examine its 1-year trend by shedding light on how potential antecedents predict its intercept and trajectory.

By this, we contribute to the nuanced understanding of STARA Awareness and the development of a comprehensive theory that advances research focusing on how humans perceive technologies in an increasingly digitalized world of work. Following the suggestions by Yam et al. (2023), we do not only consider "robots with physical forms" (Yam et al. 2023, 851) but STARA in their scope. By separating STARA Awareness from JI and identifying potential antecedents, we contribute to the development of interventions to specifically reduce STARA Awareness and its negative impacts on employees. We provide an adapted, valid questionnaire for use in research and practice that captures a digitalization-specific form of job insecurity in terms of employees' affective appraisal (concern, worry, anxiety) of STARA-driven task and job substitution. Thus, we tackle the replacement of the human workforce, which represents a central risk of new technologies from a human factors and ergonomics perspective (Salmon et al. 2023).

1 | STARA Awareness

Originally, Brougham and Haar (2018) introduced the construct of STARA Awareness. Later labeled as *threat of technological disruption*, Brougham and Haar (2020) extended its definition by the inclusion of automation and redefined the construct as "an employee's appraisal of technology (e.g., threat of smart technology, artificial intelligence, automation, robotics, and algorithms) as potentially affecting their current work through radical technological changes" (p. 3). Even though STARA Awareness represents a timely construct with essential practical implications, we suggest refinements on the following two theoretical critiques. First, although the term Awareness suggests that especially cognitive aspects of STARA are covered, the questionnaire developed by Brougham and Haar (2018, 2020) mainly includes items that capture affective elements. Taking a look at the proposed questionnaire, it encompasses three items explicitly assessing affective appraisal (e.g., "I am personally worried that what I do now in my job will be able to be replaced by STARA") (Brougham and Haar 2018, 246) and only one item that aims at the perceived possibility of job substitution due to increasing implementation of STARA ("I think my job could be replaced by STARA," Brougham and Haar 2018, 246). Therefore, this questionnaire mainly seems to capture a STARA-driven form of job insecurity in the modern world of work (Gödöllei and Beck 2023) without differentiating cognitive and affective aspects, as it is established in JI literature. The focus on the affective component raises serious questions about the fit of the construct definition and the developed measurement instrument. Going forward, the derivation of hypotheses considers this affective component that is strongly focused on the developed questionnaire by Brougham and Haar (2018). Second, both definitions provided by Brougham and Haar (2018, 2020) neglect that innovative technologies have the potential to substitute core tasks within a job which in turn causes fundamental changes in tasks instead of substituting whole jobs (Dengler and Matthes 2018).

In addition to criticism concerning its definition and measurement, an overarching theoretical model is missing so far. Brougham and Haar (2018) stated that STARA Awareness is theoretically based on the career planning literature (Greenhaus and Kopelman 1981) and the boundaryless career concept (Arthur and Rousseau 2001). STARA could potentially automate specific occupations and lead to technological disruption and downsizing of a variety of sectors (Dengler and Matthes 2018; Frey and Osborne 2017). Therefore, STARA will increasingly affect the individual's career planning process by altering workplaces and employment settings (Brougham and Haar 2018). Although a thorough construct validation of STARA Awareness is missing, several scholars already adapted versions like *AI Awareness* (Kong et al. 2021) or *automation-related job insecurity* (Gödöllei and Beck 2023).

To date, few cross-sectional findings provide insights into the relationship between STARA Awareness and potential work-related outcomes. For example, STARA Awareness correlates positively with burnout (Brougham and Haar 2018; Kong et al. 2021) and turnover intentions (Brougham and Haar 2018, 2020; Gödöllei and Beck 2023; Li et al. 2019), and negatively with organizational commitment, career

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satisfaction (Brougham and Haar 2018) and work engagement (Gödöllei and Beck 2023).

The correlations reported in the literature highlight the relevance of the construct. However, all these correlations have also been reported for theoretically related constructs, such as JI (Brougham and Haar 2018; Greenglass and Burke 2000; Staufenbiel and König 2011). Brougham and Haar (2018) also provided first empirical results that highlight its incremental validity for job (organizational commitment, career satisfaction, turnover intentions) and well-being outcomes (depression and cynicism) beyond JI. Overall, findings suggest detrimental relations of STARA Awareness with varying outcome variables, while researchers completely neglected the investigation of antecedents. Therefore, we aim not only to provide insights into the nature of this construct by separating it from related constructs but also to suggest a specific theoretical framework of JI (Shoss 2017) that helps to identify and test potential antecedents of it. Insights about the separation from job insecurity and antecedents are needed to identify risk groups experiencing STARA Awareness and to develop novel interventions to specifically reduce STARA Awareness and counteract its detrimental downstream effects.

1.1 | Differentiating STARA Awareness from Job Insecurity

JI is defined as "the degree to which employees perceive their jobs, or important features of their jobs, to be threatened and to which they perceive themselves to be powerless to do anything" (Huang et al. 2010, 21). Research on JI generally differentiates two highly intercorrelated, but separate factors ($r = 0.53$, Jiang and Lavaysse 2018), namely *cognitive* and *affective JI* (Huang et al. 2012). Cognitive JI captures the "perception of the likelihood of negative changes to one's job (e.g., losing the job or losing attractive job features), while affective JI captures the affective elements of the JI experience, such as being concerned, worried, or anxious about losing the job or particular job features" (Huang et al. 2010, 22). Affective JI is considered to be the consequence of cognitive JI and to mediate the effects of cognitive JI on outcomes like burnout (Jiang 2024; Jiang and Lavaysse 2018). This represents the most-studied separation of JI factors (Jiang and Lavaysse 2018).

Other differentiations distinguish between *quantitative JI* (perceived threat to one's job) and *qualitative JI* (perceived threat of valued job features) (Shoss 2017). The debate about whether they are parallel stresses or whether quantitative JI is the consequence of qualitative JI is ongoing (Jiang 2024). Klug et al. (2024) classify all forms of JI as hindrance stressors with negative effects on employees' health, well-being, performance, and motivation. Although all forms of job insecurity share specific core elements, such as the perceived threat of losing one's job or valued job features and a perceived uncertainty about future employment (Klug et al. 2024), these separations are widely accepted, empirically supported, and essential to understand the complex phenomenon of JI (Jiang 2024). Even integrated models exist that combine cognitive and affective, quantitative and qualitative dimensions of JI, but are not yet established (Klug et al. 2024).

Given the striking similarity of the STARA Awareness construct with already existing constructs, in particular JI, it is surprising that—to our knowledge—no prior research has provided empirical insights if these constructs can be separated. On the one hand, STARA Awareness and JI align in the content through a perceived threat to the current job as well as uncertainty about future employment (Brougham and Haar 2020; Huang et al. 2010). This seems to be reflected in prior empirical findings, demonstrating (partly) strong intercorrelations of $r = 0.51$ (Brougham and Haar 2020), or $r = 0.71$ (Lingmont and Alexiou 2020). On a theoretical level, some researchers consider STARA Awareness as an antecedent of JI which mediates the effects of STARA Awareness on work outcomes (Brougham and Haar 2020; Lingmont and Alexiou 2020), similar to cognitive JI influencing work outcomes, mediated by affective JI.

On the other hand, STARA Awareness includes a digitalization context in its definition and measurement (STARA substituting work processes, therefore leading to downsizing in organizations and industries), whereas JI is defined in general terms (causes of job insecurity are not explicitly stated in definition and measurement). Additionally, the two constructs differ in terms of their abstraction level concerning one's job or organization and industry. While STARA Awareness is operationalized by the substitution of one's job in one's organization and industry, JI only considers an individual's perceived threat of the current job (Brougham and Haar 2018). Given these differences, empirical findings also provide initial hints that STARA Awareness can be separated from JI. In their first study, Brougham and Haar (2018) did not find a significant correlation in a sample of 120 employees.

Additionally, in prior studies on STARA Awareness, JI was operationalized as a global construct and not differentiated into cognitive and affective JI, thus it is not clear yet if both JI aspects are comparably related to it. Given the substantial construct overlap in the content of STARA Awareness (capturing both cognitive and affective components), cognitive JI, and affective JI, we postulate that they are positively related, yet differentiated empirically.

Hypothesis 1. *A three-factor model separating STARA Awareness, cognitive and affective JI fits better than a one-factor model subsuming all constructs.*

Hypothesis 2. *STARA Awareness is positively related to (a) cognitive and (b) affective JI.*

1.2 | A Theoretical Framework for the Postulation of Antecedents of STARA Awareness

Shoss' (2017) *Conceptual Model of Antecedents and Outcomes of Job Insecurity*, which combines the two perspectives of psychological contract (Ashford et al. 1989) and stress research (Sverke et al. 2002), provides a detailed overview of antecedents and outcomes of JI based on comprehensive empirical findings. In this model, antecedents range from national/macro-economic factors (i.e. industry decline, technological change) to individual characteristics

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(i.e. *core self-evaluations*, CSE). Given the conceptual overlap of the two constructs, we suggest Shoss' (2017) model as a theoretical framework to also explain antecedents of STARA Awareness. We examine potential antecedents on different levels according to this model: On the national/macro-economic level, we investigate the substitution potential of an occupation (the relative share of tasks within an occupation that can be automatically performed either by a computer-controlled machine or a computer algorithm) (Dengler and Matthes 2018) and the use of STARA as indicators for the technological change. Both directly increase by implementing STARA in workplaces. On the individual level, we consider CSE as the only listed personality trait in Shoss' (2017) model to investigate whether STARA Awareness also depends on personal vulnerabilities.

First, at the national/macro-economic level, the model postulates that industry decline, shrinking demands, and technological change trigger employees' JI by generating changes that jeopardize jobs and act as distant indicators of future vulnerabilities (Shoss 2017). Indirect empirical support for this anticipated link stems from a study by Roskies and Louis-Guerin (1990) who showed that perceived changes in business or the technological environment significantly predict JI. Due to the increasing implementations of STARA in the modern world of work, the substitution of occupations and core tasks will lead to growing industry declines, shrinking demands, and increasing unemployment rates in specific sectors and occupations (Dengler and Matthes 2018; Frey and Osborne 2017). Therefore, we expect an increasing substitution potential of occupation to stimulate job-threatening changes and to be associated with higher STARA Awareness.

Hypothesis 3. *The substitution potential of occupation is positively related to STARA Awareness.*

We theorize that the use of STARA acts as a distal warning sign for the growing substitution of occupations, hence fostering STARA Awareness. By using STARA in their jobs, employees recognize the potential of STARA to outperform and substitute human workers in specific work tasks. STARA can already outperform human workers in a whole variety of cognitive and manual tasks and is "poised to outperform them in the near future" (Yam et al. 2023, 851). Additionally, organizations that implement STARA recognize its potential to substitute human work. Thus, the use of STARA can act as a proxy variable for the willingness of organizations to automate work processes, resulting in higher substitution risks and in turn higher STARA Awareness for employees who use STARA in their daily work compared to those who do not. Partial empirical support for this stems from recent findings by Yam et al. (2023) who demonstrated that the exposure to robots increases employees' JI.

Hypothesis 4. *Use of STARA is positively related to STARA Awareness.*

Second, at the individual level, Shoss' (2017) model posits CSE—a higher-order personality trait comprised of *neuroticism*, *self-esteem*, *self-efficacy*, and *locus of control* (Judge 2009)—as an antecedent of JI. Low CSE, characterized by low emotional stability, low self-esteem, low self-efficacy, and high external locus of control, promotes JI. The tendency to view oneself as unable and worthless, to evaluate events as negative and uncontrollable, and to perceive one's environment as threatening triggers JI (Shoss 2017). Based on Hobfoll's (1989) *conservation of resources theory*, Hsieh and Huang (2017) consider high CSE to act as a personal resource that equips individuals with higher capacities to cope with stressful events (e.g., the potential of job loss due to STARA), resulting in lower STARA Awareness. In a meta-analysis, Jiang et al. (2021) corroborated an average correlation of CSE with JI of $r = -0.42$. Based on this negative association between CSE and JI, we hypothesize that individuals with low CSE will have high STARA Awareness, perceiving the introduction of STARA and its impact on the future of their individual jobs as negative, threatening, and uncontrollable.

Hypothesis 5. *CSE is negatively related to STARA Awareness.*

2 | Pilot Study: Validation of a German STARA Awareness Scale

In our cross-sectional pilot study with German employees, we translate the STARA Awareness questionnaire by Brougham and Haar (2018) into German and validate it by investigating its internal structure, separating it from cognitive and affective JI (H1-H2), and identifying potential antecedents of STARA Awareness based on Shoss' (2017) JI model (H3–5).

2.1 | Methods

2.1.1 | Procedure and Participants

We invited German employees from the authors' and two student assistants' personal and professional networks from various branches and organizations via e-mail and social media accounts to an online survey. Respondents had to work at least part-time to participate. We offered two incentives in return for participation (summary of the study results, opportunity to win one of five 20€ vouchers). We measured the included variables in randomized order—except for the substitution potential that was measured at the end of the questionnaire. During the data collection from March to April 2021, 219 participants completed the study. Four participants were excluded from the analysis because they failed the built-in attention check ("Please tick answer option 1 'strongly disagree' in this row"). The final sample consisted of $N = 215$ participants (63% female). On average, they were 31.53 years old (SD = 9.23). While 49% of the respondents reported a university degree as their highest level of education, 30% reported an apprenticeship, resulting in a disproportionately highly academic German sample. The participants stemmed from various branches (22% from manufacturing, 14% from healthcare, 8% from science), including occupations with strongly varying substitution potentials and use of STARA (see Table 1 in ESM for an overview of represented industries).

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2.1.2 | Measures

2.1.2.1 | STARA Awareness. STARA Awareness was measured using the 4-item scale by (Brougham and Haar 2018). Two independent researchers translated the scale into German. Accuracy was confirmed through back translation by a native English speaker. We adapted the items following the development of the construct suggested by Brougham and Haar (2020) by including the word "automation" in the items. However, we left algorithms out as they represent the basis for smart technologies, artificial intelligence, robotics, and automation. One exemplary adapted item is "I am personally worried about my future in my organization due to smart technology, artificial intelligence, robotics and automation replacing employees" (original item: I am personally worried about my future in my organization due to smart technology, artificial intelligence, robotics and algorithms replacing employees). Items were answered as for the original scale on a five-point Likert scale ranging from *completely disagree* (1) to *completely agree* (5). Internal consistency was good (Cronbach's $\alpha = 0.87$, McDonald's $\Omega = 0.87$).

2.1.2.2 | Job Insecurity (JI). Cognitive and affective JI were measured using a validated German translation (Staufenbiel and König 2011) of the job insecurity scales by Borg and Elizur (1992). Cognitive JI was measured with four (e.g., "My job is secure"), affective JI with three items (e.g., "The thought of losing my job troubles me"). Responses were made on a scale from *strongly disagree* (1) to *strongly agree* (7). Affective JI demonstrated very high internal consistency ($\alpha = 0.93$, McDonald's $\Omega = 0.93$), whereas cognitive JI showed high internal consistency ($\alpha = 0.85$, McDonald's $\Omega = 0.86$).

2.1.2.3 | Substitution Potential. The substitution potential of the occupation of participants was determined using the Job Futuromat website by the German Institute for Employment Research (https://job-futuromat.iab.de). The substitution potential is based on task profiles of occupations and indicates which and how many core tasks of the occupation could be performed fully automatically by a computer or a computer-controlled machine as of 2019 for a selected occupation. The substitution potential presents the relative number of core tasks of occupation that can be automated and varies between zero and 100 percent. As the system differentiates various jobs within occupational types, we instructed participants to open the website and enter the accurate job title for their current position.

2.1.2.4 | Use of STARA. Use of STARA was assessed with a single item ("Please estimate the percentage of your weekly work time in which you have contact with smart technology, artificial intelligence, robotics, and automation"). Respondents answered on a scale of zero to 100 percent in steps of 10 percent.

2.1.2.5 | CSE. CSE was measured with the validated German version (Stumpp et al. 2010) of the twelve-item scale originally developed by Judge et al. (2003). One exemplary item was "When I try, I generally succeed." Participants answered on a five-point Likert scale ranging from *strongly disagree* (1) to *strongly agree* (5). The overall scale yielded good reliability ($\alpha = 0.86$, McDonald's $\Omega = 0.86$).

2.1.3 | Analytic Approach

We performed all statistical analyses using JASP 0.17.1 (JASP 2023). For the measurement models, we compared a *one-factor model* to a *three-factor model*, using confirmatory factor analyses (CFA). In the one-factor model, all items (measuring STARA Awareness, cognitive JI, and affective JI) load onto one general factor. The three-factor model represents three different, intercorrelated factors, namely STARA Awareness, cognitive JI, and affective JI. We based our evaluations of the model fit on the recommendations by Hooper et al. (2008) and West et al. (2012) by using the comparative fit index (CFI > 0.95), standardized root-mean-square residual (SRMR < 0.08). Moreover, we report the Akaike information criterion (AIC). We also report the average variance extracted (AVE) of STARA Awareness and its shared variance (SV) with cognitive and affective JI. AVE scores that exceed SV scores support discriminant validity (Farrell 2010).

2.2 | Results and Discussion

Table 1 lists the results of the measurement model of STARA Awareness (see factor loadings in Figure S1 in ESM), and the two CFAs comparing STARA Awareness and cognitive and affective JI. CFI and SRMR suggest a good fit of the STARA Awareness measurement model for STARA Awareness. However, the item containing cognitive components exhibits the lowest factor loading. In line with our expectations, a three-factor model separating STARA Awareness, cognitive and affective JI fits better than the one-factor model, thereby supporting H1.

TABLE 1 | Fit indices of the CFA models of STARA Awareness, cognitive and affective JI and average variance extracted (pilot study).

Model	χ^2	*df*	χ^2/df	$\Delta\chi^2(\Delta df)$	CFI	SRMR	AIC	AVE
STARA Awareness measurement model	17.442	2	8.72		0.965	0.037	42.442	0.620
Model Comparisons (including items for cognitive and affective job insecurity)								
One-factor model	1083.377	44	24.62		0.311	0.255	1149.377	
Three-factor model	111.770	41	2.73	971.68(3)***	0.953	0.065	183.770	

Note: $N = 215$. χ^2 difference test compares to the previous model.
Abbreviation: AVE, average variance extracted.
***$p < 0.001$.

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TABLE 2 | Descriptive statistics and bivariate Spearman intercorrelations of study variables (pilot study).

Variables		*M*	SD	1	2	3	4	5	6
1	STARA Awareness	1.83	0.80	(0.87)					
2	Cognitive JI	2.82	1.30	0.17*	(0.85)				
3	Affective JI	3.12	1.75	0.16*	0.25***	(0.93)			
4	SP	35.39	31.40	0.34***	−0.16*	−0.03	—		
5	Use of STARA	32.42	28.62	0.21**	−0.08	0.00	0.35***	—	
6	CSE	3.76	0.58	−0.12	−0.19**	−0.32***	0.04	0.11	(0.86)

Note: $N = 215$. Cronbach's alpha in bracket. Responses in SP and Use of STARA in percent.
Abbreviations: CSE, core self-evaluations; JI, job insecurity; SP, substitution potential.
*$p < 0.05$. **$p < 0.01$. ***$p < 0.001$. SP is stronger related to STARA Awareness than to cognitive JI, $z = 6.208$, $p < 0.001$, and affective JI, $z = 4.446$, $p < 0.001$. Use of STARA is stronger related to STARA Awareness than to cognitive JI, $z = 3.402$, $p < 0.001$, and affective JI, $z = 2.428$, $p = 0.015$. CSE is not stronger related to cognitive JI than to STARA Awareness, $z = -0.812$, $p = 0.417$, but stronger related to affective JI than to STARA Awareness, $z = -2.373$, $p = 0.018$.

Table 2 displays descriptive statistics and Spearman intercorrelations of the variables in the pilot study (none of the variables was normally distributed according to Shapiro-Wilk tests, $p < 0.001$). In line with our expectations, STARA Awareness was positively related to cognitive and affective JI in a comparable height, supporting H2a and H2b. Since AVE (0.620) of STARA Awareness exceeded its SV with cognitive (0.029) and affective JI (0.026), the analyses support the discriminant validity of STARA Awareness. Furthermore, the substitution potential of occupation and use of STARA were both positively associated with STARA Awareness, supporting H3 and H4. CSE and STARA Awareness were not significantly related, $r = -0.12$, $p = 0.07$, rejecting H5. In contrast to STARA Awareness, we did not find a significant relationship between substitution potential and affective JI, $r = -0.03$, $p = 0.63$, or between the use of STARA and cognitive JI, $r = -0.08$, $p = 0.22$, or affective JI, $r = 0.00$, $p = 0.97$. Substitution potential and cognitive JI were even negatively related. CSE was negatively related to cognitive and affective JI (see Table 2 in ESM for an overview of results across all conducted studies).

In our pilot study, we successfully translated the existing STARA Awareness measurement instrument by Brougham and Haar (2018). Similar to the studies by Brougham and Haar (2018, 2020), STARA Awareness is low in magnitude in our German sample. We separated STARA Awareness from cognitive and affective JI on the measurement level, even though they are weakly related to one another. The relationships between STARA Awareness and cognitive and affective JI are much weaker than in prior studies that examined JI globally (Brougham and Haar 2020; Lingmont and Alexiou 2020). Even though STARA Awareness is largely affective in nature, it is similarly related to cognitive and affective JI.

With regard to potential antecedents, we demonstrated most of the expected correlations based on Shoss' (2017) model. Although these findings seem to support that the model can be transferred to STARA Awareness, comparing the correlation pattern of this construct with (cognitive and affective) JI indicated substantial differences. Compared to JI, STARA Awareness seems to be more strongly associated with national/macro-economic factors than with personality factors, such as CSE. That STARA Awareness rises with increasing use of STARA suggests that fears of job loss due to the implementation of STARA at work are not reduced by the mere use of these technologies. Rather, this is a growing concern in an increasingly digitized work environment. Hence, STARA Awareness represents a distinct construct with its own nomological net.[1] Therefore, our study provides valuable information for the theoretical and practical value of the STARA Awareness construct.

However, given the mismatch of the cognitive framing of the STARA Awareness term and definition and its measurement with largely affective items, we suggest the need for a revision of the original construct and questionnaire. The affective items assess concerns about the substitution of jobs on varying levels (individual, organizational, and industrial level), whereas the cognitive item only captures the perceived possibility of the substitution of one's job at the individual level. However, the introduction of STARA in the modern world of work will not only lead to the complete substitution of jobs but also to the substitution of individual tasks and thus to job-related changes (Dengler and Matthes 2018), which is neglected in the definition and measurement of STARA Awareness (Brougham and Haar 2018, 2020). Based on the suggested theoretical overlap and our findings, we propose the renaming of the construct and its consideration as a digitalization-specific form of JI, which has also been suggested by Gödöllei and Beck (2023). Following the call by Jiang (2024) to differentiate cognitive and affective aspects of JI and reach construct clarity, we suggest a revision of the (original) questionnaire in accordance with research on JI. Based on the perspective of distinguishing cognitive and affective JI, we define and aim to assess cognitive and affective components of automation-related JI separately. We define *cognitive automation-related job insecurity* (CAJI) as the employees' perceived possibility of STARA-driven task and job substitution. Relatedly, we define *affective automation-related job insecurity* (AAJI) as employees' affective appraisal (concern, worry, anxiety) of STARA-driven task and job substitution.

3 | Study 1: Separation of Cognitive and Affective Automation-Related JI

In this cross-sectional study with German employees, we address the aforementioned content criticism of the construct. We expand the limited STARA Awareness construct by adding cognitive and affective items that capture components at the individual, organizational, and industrial level, and substitution

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of tasks within jobs. We investigate whether we can separate STARA Awareness into CAJI and AAJI. Additionally, we aim to replicate our validation hypotheses (H1-5) from the pilot study.

Hypothesis 6. *A two-factor model of CAJI and AAJI fits better than a one-factor model.*

3.1 | Methods

3.1.1 | Procedure and Participants

Again, we invited German employees from our and one student assistant's personal and professional networks from various sectors and organizations for a short cross-sectional online study. They had to work at least part-time to participate. Again, we rewarded the participants with a summary of the central study results and the opportunity to win one of five 20€ vouchers. As in the pilot study, we assessed the variables in randomized order but collected the substitution potential at the end of the questionnaire. During 3 weeks in October 2021, 227 participants completed this study. We excluded one person who reported "student" as current job, one person who reported having simply looked through the questionnaire, and another one for working under 16 h per week, leaving 224 participants. The majority of our sample was female (58%). On average, the participants were 33.08 years old (SD = 10.26). 63% of the participants had a university degree, whereas 23% completed an apprenticeship, again resulting in a sample with an above-average number of academics. Participants worked in various branches (see Table S1 in ESM).

3.1.2 | Measures

We assessed STARA Awareness, cognitive and affective JI, the substitution potential, the use of STARA, and CSE with the same methods used in the pilot study.

3.1.2.1 | Cognitive and Affective Automation-Related Job Insecurity. To extend the construct, we followed the recommendations by Hinkin (1998) and Schwab (1980). One author generated 26 new items based on the definitions of CAJI and AAJI and the aforementioned content-related criticism on the STARA Awareness construct (see Figure S2 in ESM for a visualization of the item adaption process). After screening our generated items regarding redundancy and representativeness (Ferris et al. 2008), another author excluded seven items, resulting in twelve cognitive and seven affective items that represented automation-related job insecurity on the individual, organizational, and industrial level, and the substitution of individual tasks. Following recommendations by Clark and Watson (1995), the original and generated items were evaluated by two research associates and one research assistant for how close they fit the construct definition of CAJI and AAJI on a five point Likert-scale from *strongly disagree* (1) to *strongly agree* (5). Based on these ratings, we excluded six cognitive and three affective items, resulting in the four original (one cognitive and three affective items), six new cognitive, and four new affective items. We tested these 14 items empirically in Study 1 (see item formulation and factor loadings in Table 3 in ESM).

3.1.3 | Analytic Approach

We performed all statistical analyses including exploratory factor analyses (EFAs), CFAs for the original (STARA Awareness), and its adapted measurement scale using JASP 0.17.1 (JASP 2023). We report AVE and SV scores.

3.2 | Results and Discussion

All model fit indices of the measurement model of the original STARA Awareness scale suggest a very good model fit (Table 3, see factor loadings in Figure S3 in ESM). Considering our criticism of its content, we adapted the scale by Brougham and Haar (2018) to reach content clarity (Jiang 2024). Therefore, we conducted an EFA using maximum likelihood and promax rotation (Cureton and Mulaik 1975) with 14 items, which suggested the extraction of two factors, preliminarily supporting H6. However, the factor loadings on these two factors did not show any specific pattern. We did not find evidence that items capturing the perceived possibility of STARA-driven task and job substitution and items assessing affective appraisal of STARA-driven task and job substitution loaded on separate factors. Contrary to our expectations, all items except three cognitive items showed substantial estimated factor loadings on factor 1 (factor loadings > 0.4), whereas one inverted affective item did not show a substantial estimated factor loading on either of the two factors. To further investigate the factor structure and potential separation of CAJI and AAJI, we contrasted a one-factor model to a two-factor model with all 14 items (Table 3). In the one-factor model (see factor loadings in Figure S4 in ESM), all seven cognitive and seven affective items loaded on a single factor, while in the two-factor model (see factor loadings in Figure S5 in ESM), the cognitive items and the affective loaded on separate factors. Contrary to H6, the two-factor model did not fit significantly better than the one-factor model ($\Delta\chi^2(1) = 1.09$, $p = 0.30$). All model fit indices suggest a poor model fit for both models, including all 14 items. To improve model fit, we reduced the number of cognitive and affective items. Based on item statistics and content consideration, an alternative model including four items each did not lead to a satisfactory fit for two separate scales to assess CAJI and AAJI (see factor loadings in Figures S6 and S7 in ESM). Hence, we failed to differentiate factors assessing cognitive and affective components empirically.[2] Consequently, these results indicate that employees do not differentiate between the perceived possibility and affective appraisal (concern, worry, and anxiety) about STARA-driven task and job substitution.

However, we adapted the original four-item scale to address the content criticism and construct clarity. We removed the cognitive item but added an affective item that captures the substitution of individual tasks. Additionally, we replaced one affective item with a new item so that the item stem varied stronger. The model fit indices of the measurement model of our new scale to capture AAJI, solely focusing on affective elements encompassing four items, suggested a very good model fit (Table 3, see factor loadings in Figure S8 in ESM). Table 3 in ESM displays the items of the adapted questionnaire in German and English, with the final items bolded. Given minor changes, the original scale by Brougham and Haar (2018)

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TABLE 3 | Fit indices of the CFA models of STARA Awareness, cognitive and affective STARA Awareness, cognitive and affective JI and average variance extracted (Study 1).

Model	χ^2	*df*	χ^2/df	$\Delta\chi^2(\Delta df)$	CFI	SRMR	AIC	AVE
STARA measurement model	0.435	2	0.218		1.000	0.005	1767.049	0.652
14 Items Model comparison (including items for cognitive and affective STARA Awareness)								
One-factor model	501.581	77	6.514		0.794	0.095	7235.522	
Two-factor model	412.133	76	5.423	1.091(1)	0.837	0.085	7148.074	
AAJI measurement model	1.895	2	0.948		1.000	0.010	1660.406	0.677
Model Comparisons (AAJI and cognitive and affective JI)								
One-factor model	1143.900	44	25.793		0.339	0.264	7712.084	
Three-factor model	49.960	41	1.22	1084.94(3)***	0.995	0.035	6633.144	

Note: $N = 224$. χ^2 difference test compares to the previous model.
Abbreviations: AAJI, affective automation-related job insecurity; AVE, average variance extracted; STARA, STARA Awareness.
***$p < 0.001$.

TABLE 4 | Descriptive statistics and bivariate Spearman intercorrelations of study variables (Study 1).

Variables		*M*	SD	1	2	3	4	5	6
1	AAJI	1.62	0.71	(0.89)					
2	Cognitive JI	2.61	1.35	0.14*	(0.87)				
3	Affective JI	2.95	1.73	0.15*	0.33***	(0.94)			
4	SP	32.22	27.89	0.35***	−0.24***	−0.06	—		
5	Use of STARA	27.63	25.96	0.33***	−0.09	0.04	0.28***	—	
6	CSE	3.78	0.63	−0.15*	−0.25**	−0.32***	0.07	0.08	(0.87)

Note: $N = 224$. Cronbach's alpha in brackets. Responses in SP and Use of STARA in percent.
Abbreviations: AAJI, affective automation-related job insecurity; CSE, core self-evaluations; JI, Job insecurity; SP, substitution potential.
*$p < 0.05$. **$p < 0.01$. ***$p < 0.001$. SP is stronger related to AAJI than to cognitive JI, $z = 7.568$, $p < 0.001$, and affective JI, $z = 5.039$, $p < 0.001$. Use of STARA is stronger related to AAJI than to cognitive JI, $z = 5.124$, $p < 0.001$, and affective JI, $z = 3.503$, $p < 0.001$. CSE is not stronger related to cognitive JI than to STARA Awareness, $z = -1.179$, $p = 0.238$, but stronger related to affective JI than to AAJI, $z = -2.051$, $p = 0.040$.

and our revised scale are highly intercorrelated[3], $r = 0.92$, $p < 0.001$.

Table 3 also lists the results of the two CFAs comparing AAJI, and cognitive and affective JI. In line with our expectations, the three-factor model fits significantly better than the one-factor model. All model fit indices of the three-factor model suggest a very good model fit, thereby supporting H1 for the AAJI scale. Moreover, the AVE score of AAJI (0.677) exceeds the SV with cognitive (0.020) and affective JI (0.023), indicating the discriminant validity of AAJI.

Table 4 lists descriptive statistics and Spearman intercorrelations of the variables in Study 1 (all variables were not normally distributed as assessed by Shapiro-Wilk tests, $p < 0.001$). Supporting H2a and H2b, AAJI was positively associated with cognitive and affective JI. In line with our expectations in H3 and H4, the substitution potential of occupation and use of STARA were positively related to AAJI. Contrary to the pilot study, CSE and AAJI were negatively associated, supporting H5. In line with the pilot study, the substitution potential was not significantly related to affective JI, $r = -0.06$, $p = 0.39$, while the use of STARA was not related to cognitive JI, $r = -0.09$, $p = 0.18$, and affective JI, $r = 0.04$, $p = 0.55$. Again, the substitution potential and cognitive JI were negatively related. Additionally, CSE was negatively associated with cognitive and affective JI.

We adapted the STARA Awareness construct and questionnaire by Brougham and Haar (2018) addressing the aforementioned content criticism. Although we failed to separate aspects of CAJI and AAJI, we provide a revised, validated scale assessing employees' affective appraisal of STARA-driven task and job substitution.

Furthermore, we have almost completely replicated our previous results on potential antecedents. AAJI was more strongly related to national/macro-economic factors than JI. This suggests that the continuing digitalization of the modern world of work evokes AAJI (not general JI), underlining its additional value. In contrast to the pilot study and in line with JI, we identified CSE as a potentially protective personality trait against AAJI. This relationship probably emerged due to the now purely affective nature of our revised measurement scale. Since our two studies so far are limited by their cross-sectional design, we need longitudinal study designs to empirically test whether the potential antecedents predict the starting point and the trajectory of the construct.

4 | Study 2: The Stability and Change of Affective Automation-Related Job Insecurity

We conducted a longitudinal study over 1 year with three measurement points and time lags of 6 months each, beginning

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in January 2022 and ending in March 2023. Our study aim was threefold. First, we cross-validate our AAJI scale. Second, we investigate its 1-year trend. Third, we provide insights into how potential antecedents predict the starting point (intercept) and trajectory (slope) of AAJI over time. Besides the postulated hypotheses 3–5 that we tested in prior studies, we propose and test two further hypotheses in Study 2.

As in the prior studies, we anticipate that the substitution potential of occupation and the use of STARA are positively and CSE are negatively related to AAJI, thereby determining the starting level (intercept) of AAJI at T1.

Hypothesis 7. *Substitution potential, use of STARA and CSE will predict AAJI at the starting point of the study (intercept) such that higher scores on substitution potential (H7a) and use of STARA (H7b), and lower scores on CSE (H7c) are associated with higher initial AAJI levels.*

Further, we postulate that the use of STARA does not only influence the intercept but also a positive growth trajectory of AAJI. The acquisition of knowledge, abilities, and skills by humans is naturally limited. In contrast, the rapidly increasing technological advances of STARA widen the discrepancies between human and technological performance in favor of the latter. This also applies to industries and occupations, in which STARA has not yet surpassed human performance (Yam et al. 2023). Based on the natural human limit and resulting limited coping opportunities (in terms of acquisition of knowledge, abilities, and skills), we postulate that employees increasingly recognize STARA's potential to substitute tasks and jobs by using STARA at work. This may result in rising AAJI over time.

Hypothesis 8. *Use of STARA will predict positive growth in AAJI (slope) such that higher scores on the use of STARA are associated with higher positive growth in AAJI.*

4.1 | Methods

4.1.1 | Open Science

We pre-registered all procedures and hypotheses before data collection (https://doi.org/10.17605/OSF.IO/CUGVK).

4.1.2 | Procedure and Participants

We invited German employees from various sectors and organizations via e-mail and social media accounts for a short longitudinal online study with three measurement points over 1 year with a time-lag of 6 months, similar to longitudinal studies on antecedents of JI (Huang et al. 2012; Kinnunen et al. 2014). The participants had to work part-time at least. We rewarded participants with evidence-based tips for remote work, individual feedback on their work motivation (assessed at T1), and the chance to win one of 20 vouchers (10€). We donated 1€ to a charitable organization for every complete participation. We collected the substitution potential at the end of the questionnaire from T1.

From January to March 2022, 498 participants completed T1 and were automatically invited to the following measurement point 6 months later. We excluded 34 participants who failed the built-in attention check ("Please tick answer option 5 'strongly agree' in this row."), eight participants who worked less than 16 h per week, four participants who stated "student" as their current job, and four participants with missing values in the substitution potential from the statistical analysis. Furthermore, we excluded participants who changed jobs ($n = 41$) between T1 and T3, resulting in 406 participants that we included in T1. Of the 406 participants at T1, 261 completed T2 and were automatically invited to the final measurement point (T3) 6 months later. Of the 261 participants from T2, 233 participants also completed T3. To investigate potential systematic drop-out we tested for differences in study variables (AAJI, substitution potential, use of STARA, and CSE) in participants who completed all three measurement points ($N = 233$) from participants who did not participate in T2 and/or T3 ($N = 173$) using MANOVA. Results indicated no systematic dropout based on our study variables, $F(4, 401) = 1.21$, $p = 0.31$.

The final sample included 117 women (50%), 114 men (49%), one gender-diverse participant, and one participant who did not state gender. The average age was 37.31 years (SD = 12.13). With 58% of participants with a university degree and 26% with an apprenticeship as their highest level of education, the sample is disproportionately highly academic. Again, participants stemmed from various branches (see Table 1 in ESM) and occupations.

4.1.3 | Measures

We assessed the substitution potential, the use of STARA, and CSE with the identical methods used in the prior studies. We used our modified scale from Study 1 to measure AAJI.

4.1.4 | Analytical Approach

We tested the measurement model of the AAJI scale at all three measurement points. To test H7-H8, we performed latent growth curve models (LCM) using R and the package lavaan (Rosseel 2012). Contrary to basic regression models with fixed effects which postulate that participants do not differ in their starting point (intercept) or change rate (slope), LCM considers that participants vary in their intercept and slope by adding random-effects terms (Curran et al. 2010). By performing LCM, we investigated the relationship of the three potential antecedents (substitution potential of occupation, use of STARA, and CSE) and AAJI at the starting point of the study (intercept) and the growth trajectory (slope) of AAJI over the three measurement points. For this, we followed the recommendation by Curran et al. (2010). First, we test a baseline model (M0) and evaluate its model fit. This baseline model postulates a linear positive trajectory of AAJI without the between-person predictor variables and incorporates fixed and random effects. The fixed effects include the estimation of mean intercept and mean slope. Random effects include estimates of the variance in intercept and slope, "referring to between-person variability

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around the mean intercept and mean slope" (Apers et al. 2019, 220). Second, we test a second model (M1) that incorporates between-person predictor variables at T1 (substitution potential of occupation, and CSE) to predict the individual variability in the intercept and use of STARA to also predict the individual variability in the slope of AAJI (Apers et al. 2019; Curran et al. 2010). Since scalar measurement invariance is the requirement for conducting LCM, we also analyzed longitudinal measurement invariance on different levels (configural, metric, scalar) and reported (differences in) CFI, RMSEA, SRMR. For the assessment of measurement invariance level, we follow the recommendations by Chen (2007) for small samples ($N < 300$): "a change of ≤ -0.005, supplemented by a change of ≥ 0.010 in RMSEA or a change of ≥ 0.025 would indicate non-invariance" (Chen 2007, 501).

4.2 | Results and Discussion

Table 4 in the ESM displays the measurement model of AAJI at the three measurement points and measurement invariance analyses. CFI and SRMR indicate a good model fit of AAJI at all three measurement points (see factor loadings in Figures S9–11 in ESM). In addition, the results demonstrated scalar measurement invariance for AAJI ($\chi^2 = 56.634$, CFI = 0.978, ΔCFI = −0.007, RMSEA = 0.096, ΔRMSEA = 0.000, SRMR = 0.040, ΔSRMR = 0.008), representing a reasonably good fit. Thus, the relationship between the latent variable and its indicators is consistent over time as factor loadings are equal and intercepts of observed variables are consistent across the three time points, fulfilling the prerequisite for LCM (Chen 2007). Table 5 shows descriptive statistics and Spearman intercorrelations of the variables in Study 2 (none of the variables was normally distributed according to Shapiro-Wilk tests, $p < 0.05$). Since we identified significant positive relations between the substitution potential of occupation and the use of STARA with AAJI at all three measurement points, our results support H3 and H4. In line with Study 1, we also found significant negative relations between CSE and AAJI at all three measurement points, supporting H5.

Table 6 displays the LCM results for the baseline model (Model M0) and the model including predictors for the intercept and slope (Model M1). M1 provides the basis for testing H7-8. Model fit indices for M0 suggested an acceptable fit to the data with the exception of RMSEA, Normed $\chi^2 = 6.817$, $df = 1$, $p = 0.009$, CFI = 0.980, TLI = 0.941, RMSEA = 0.158, SRMR = 0.031, AIC = 1298.361. Thereby, the results indicate that AAJI is constant both between participants and over time. M0 revealed no linear change in AAJI throughout the three measurement points (Figure S12 in ESM). The mean intercept which displays the average AAJI level at T1 was $b = 1.688$, $t(1) = 32.791$, $p < 0.001$. The mean slope which displays the average rate of AAJI growth over the three measurement points was nonsignificant, $b = 0.001$, $t(1) = 0.060$, $p = 0.953$. Intercept variance was significant, $\sigma^2_{\text{intercept}} = 0.463$, $t(1) = 6.942$, $p < 0.001$. Slope variance was nonsignificant, $\sigma^2_{\text{slope}} = 0.041$, $t(1) = 1.635$, $p = 0.102$. Intercept and slope were strongly negatively related, $r = -0.49$, $p = 0.032$, indicating that the higher the starting point the lower the change in AAJI.

Including the predictors for the intercept and slope in M1 yielded a good model fit, Normed $\chi^2 = 12.705$, $df = 8$, $p = 0.122$, CFI = 0.987, TLI = 0.981, RMSEA = 0.050, SRMR = 0.029, AIC = 1230.235, allowing the interpretation of the parameter estimations. M1 revealed no linear positive trajectory of AAJI, too, $b = -0.004$, $t(1) = -0.166$, $p = 0.868$, but slope variance was significant, $\sigma^2_{\text{slope}} = 0.037$, $t(1) = 2.754$, $p = 0.006$. Intercept and slope were strongly negatively related again, $r = -0.48$, $p = 0.007$. Although all model fit indices were better on a descriptive level, M1 did not significantly differ from M0, $\chi^2(7) = 5.888$, $p = 0.533$. The inclusion of predictors for the intercept and slope revealed the substitution potential, $\beta = 0.271$, $p < 0.001$, and the use of STARA, $\beta = 0.131$, $p = 0.005$, as significant positive predictors, supporting H7a and H7b. CSE was a significant negative predictor of the intercept variability of AAJI, $\beta = -0.136$, $p < 0.001$, supporting H7. Thus, participants with higher substitution potential and use of STARA reported higher levels of AAJI at the starting point of the study. Contrary, participants with higher CSE reported lower levels of AAJI at the starting point. Although the use of STARA was a significant predictor of slope variability of AAJI, we identified a negative prediction weight, $\beta = -0.055$, $p = 0.016$, therefore rejecting H8. This indicates that the higher the use of STARA at T1, the lower the change in AAJI over the course of 1 year. Plotting the trajectory of AAJI over the course of the three measurement points for different levels of use of STARA at T1 displays a slight increase for moderate levels of use of STARA but slight decreases for both high (M + SD) and low levels (M − SD) (Figure 1).

Applying a longitudinal study design with a total time lag of 1 year, our contributions to the emerging research field are threefold. First, we successfully cross-validated our AAJI scale in another sample. All indices suggest a good model fit of the adapted scale over the three measurement points. Second, since scholars have investigated the underlying construct of STARA Awareness in cross-sectional studies, we are the first to examine the 1-year trend of AAJI. Instead of an expected linear positive growth trajectory, we provided evidence for rather high stability over the period of 1 year. The high correlations of AAJI between the three measurement points are comparable to studies on JI which used the same total time lag (Kinnunen et al. 2014), suggesting equivalent stability for the two constructs. Third, we demonstrated that the substitution potential of occupation and the use of STARA positively predicted the intercept, whereas CSE negatively predicted the intercept. Since the intercept variance was significant, we showed that participants differed in their (relatively low) starting point of their AAJI. However, taking a closer look at the change trajectories over time, we found no indication of a general – positive or negative linear – trend. Our analysis also revealed a more complex pattern, as such that change in AAJI is associated with the initial level of use of innovative technologies at work.

5 | General Discussion

We conducted three studies with the overall aim of further developing a content-valid digitalization-specific form of job insecurity by revising the STARA Awareness construct and positioning it in the JI literature. Thus, we contribute to the

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TABLE 5 | Descriptive statistics and bivariate Spearman intercorrelations of study variables at three measurement points (Study 2).

Variables			*M*	SD	1	2	3	4	5	6	7	8	9	10	11	12
T1	1	AAJI	1.71	0.79	(0.90)											
	2	SP	36.09	29.74	0.35***	—										
	3	Use of STARA	33.98	28.76	0.20***	0.23***	—									
	4	CSE	3.68	0.61	−0.25***	−0.05	0.07	(0.85)								
T2	5	AAJI	1.63	0.73	0.64***	0.45***	0.20**	−0.21**	(0.88)							
	6	SP	38.09	29.85	0.40***	—	0.25***	−0.04	0.45***	—						
	7	Use of STARA	34.56	28.43	0.30***	0.44***	0.64***	−0.03	0.34***	0.44***	—					
	8	CSE	3.72	0.63	−0.14*	0.04	0.07	0.74***	−0.17**	0.04	0.01	(0.87)				
T3	9	AAJI	1.71	0.70	0.57***	0.36***	0.18**	−0.18**	0.65***	0.36***	0.26***	−0.12	(0.88)			
	10	SP	38.78	30.17	0.36***	—	0.30***	−0.02	0.44***	—	0.46***	0.03	0.36***	—		
	11	Use of STARA	33.05	26.60	0.27***	0.33***	0.65***	0.02	0.33***	0.33***	0.70***	0.03	0.27***	0.33***	—	
	12	CSE	3.72	0.61	−0.17*	0.00	0.06	0.74***	−0.18**	0.00	0.07	0.75***	−0.19**	0.00	0.04	(0.88)

Note: $N_{T1} = 406$. $N_{T2} = 261$. $N_{T3} = 233$. Participants with changes in substitution potential (by changes in jobs) were excluded. Cronbach's alpha in brackets. Responses in SP and Use of STARA in percent.
Abbreviations: AAJI, affective automation-related job insecurity; CSE, core self-evaluations; SP, substitution potential.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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TABLE 6 | LCM results for baseline model (M0) and with predictors (M1) with criterion affective automation-related job insecurity (Study 2).

	M0		M1	
	β	SE	β	SE
Effects of predictors on intercept				
SP			0.271***	0.038
Use of STARA			0.131**	0.046
CSE			−0.136***	0.037
Effects of predictors on slope				
Use of STARA			−0.055*	0.023
Variance components				
Intercept	0.463		0.324	
Slope	0.041		0.037	
$r_{\text{intercept, slope}}$	−0.49*		−0.48**	
R^2				
Intercept			0.288	
Slope			0.074	
Model fit indices				
Normed χ^2	6.817		12.705	
CFI	0.980		0.987	
TLI	0.941		0.981	
RMSEA	0.158		0.050	
SRMR	0.031		0.029	
AIC	1298.361		1230.235	

Note: $N = 233$. Predictors were z-standardized.
Abbreviations: CSE, core self-evaluations; SP, substitution potential.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

emerging research field on how humans perceive the implementation of new technologies in four important ways. Based on the work of Brougham and Haar (2018), we adapted STARA Awareness to address the substitution of core tasks and adapted and content-validated the scale. We redefined the measurement and thereby the corresponding construct in terms of *affective automation-related job insecurity* (AAJI) as a digitalization-specific form of JI based on content criticism. Overall, our analyses indicate a good internal structure of the revised questionnaire. In sum, our adapted construct addresses how employees perceive the impact of technological changes on their employment in an increasingly digitalized world of work in a more adequate way. We also provide a short and valid questionnaire to capture AAJI for use in research and practice.

We contribute to the investigation of the nomological net of AAJI by differentiating it from related constructs of cognitive and affective JI. Therefore, AAJI represents a contextualized measure that is only weakly related to job insecurity, emphasizing that giving participants a digitalization frame alters their perception of JI.

Drawing upon the conceptual model by Shoss (2017), we investigated the substitution of occupation, use of STARA, and CSE as potential antecedents, expanding knowledge about antecedents of AAJI. Our findings support the notion that indicators for technological change play an essential role in the development of AAJI. In line with the model by Shoss (2017), we identified CSE as a protective higher-order personality trait.

Finally, we provided insights into slight changes in AAJI over 1 year. We found evidence that the substitution potential, and the use of STARA and CSE determine the starting level of AAJI. Employees who work in occupations with high substitution potential, use more STARA at work, and score low on CSE represent risk groups of AAJI. Moreover, our results indicate that the trajectory of AAJI depends on the extent to which employees use STARA at work. As employees with moderate use of STARA at work exhibit slightly increased levels of AAJI over time, they are a critical group for future interventions. Hence, our work suggests vital implications for theory and practice.

5.1 | Theoretical Implications

Our research contributes to a comprehensive theory on AAJI. Based on the original STARA Awareness construct, our findings led to the clear conceptualization and validated assessment of AAJI, that enables the development of a theoretical foundation to foster JI and human-technology interaction research.

Our results support the transfer of indicators of technological change and higher-order personality traits as antecedents postulated in the conceptual model of JI (Shoss 2017) to AAJI. That we consistently identified a positive relationship between the substitution potential of occupation and AAJI in three studies indicates that there seems to be a rational basis for employees' concern, worry, or anxiety in terms of actual risk of being substituted. Hence, employees working in occupations with high substitution potential represent a risk group of AAJI. In turn, this can undermine well-being and positive work outcomes (Brougham and Haar 2018; Li et al. 2019; Kong et al. 2021). By broadening the technologies under consideration, our work provides insights into how STARA evokes employees' concern, worry, and anxiety about STARA-driven task and job substitution. Not only the exposure to robots but also the use of STARA is related to unintended detrimental effects on employee attitudes. Following Yam and colleagues' (2023) recommendation to consider further outcomes besides JI—in our case AAJI—our work strengthens that employees tend to appraise the implementation of new technologies negatively as they fear potential job substitution. Eventually, our findings indicate the high relevance of national/macro-economic factors that depict the technological change for AAJI.

Surprisingly, besides an even negative relation between the substitution potential and cognitive JI, we did not find any correlations between the indicators of technological change and JI. This suggests that the underlying model by Shoss (2017) needs further refinements to account for the role of technological advances in the workplace and their effect on JI. In addition, the differences in the correlation patterns between

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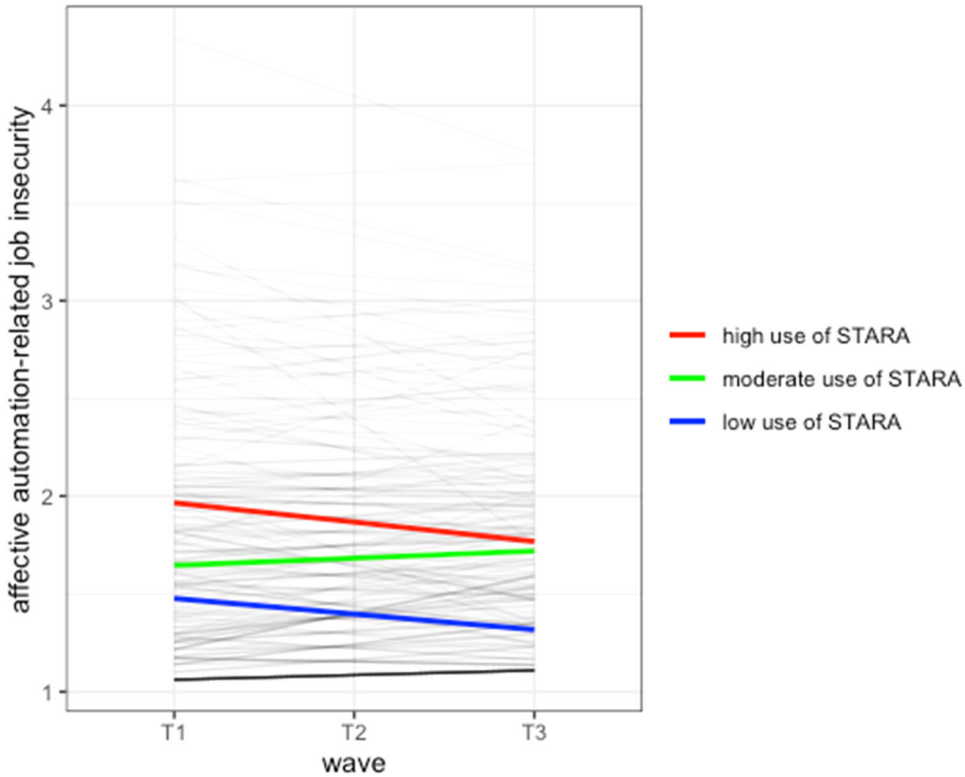

FIGURE 1 | M1: Change of affective automation-related job insecurity over the three measurement points as a function of use of STARA (Study 2). $N = 233$. The colored lines represent separate regression lines in affective automation-related job insecurity over 1 year for participants with low, moderate and high use of STARA, respectively. Gray lines represent regression lines for every participant.

JI and AAJI demonstrate that AAJI constitutes a novel construct with unique correlations with indicators of technological change.

However, AAJI and JI resemble each other in some aspects. First, CSE seem to represent a protective higher-order personality trait for the experience of cognitive and affective JI (Jiang et al. 2021), and AAJI. Therefore, our findings highlight the importance of individual differences for AAJI. Appraising oneself as worthy, competent, and capable (Stumpp et al. 2010) affects how employees experience their employment also in times of technological change. Second, AAJI exhibits comparably high stability as JI (Kinnunen et al. 2014). The slight decline of AAJI among employees with high use of STARA may be because they hold jobs like computer scientists or engineers and thus actively shape the technological change; employees who do not use STARA at work may underestimate the potential impact of STARA in the long run, also resulting in slight decreases. This further highlights the relevance of a digitalization-specific construct and measure to adequately assess employees' appraisal of the impact of new technologies on their employment.

5.2 | Practical Implications

We provide a valid questionnaire that measures AAJI, that considers that new technologies at work often have the potential to substitute single work processes and not eliminate whole jobs (Dengler and Matthes 2018). Practitioners who intend to capture employees' affective appraisal of new technologies on their employment should therefore rely on our adapted scale to assess AAJI instead of general JI scales.

Furthermore, practitioners need to consider the outlined risk groups when implementing STARA at work. Our studies suggest that CSE represent a vital higher-order personality trait in the context of personnel selection, especially in work environments that are characterized by dynamic technological advances. Although originally proposed as a trait, CSE can also represent a state, meaning "individuals with positive state CSE appraise themselves in a positive manner in a specific situation" (Nübold et al. 2013, 31). Based on findings that indicate that performance feedback triggers (state) CSE (Schinkel et al. 2004), practitioners may increase employees' state CSE and reduce AAJI by implementing frequent and immediate feedback. Practitioners should not expect decreases in AAJI by mere use or growing familiarity with new technologies over time based on different trajectories.

5.3 | Limitations and Future Research

We have to acknowledge several limitations of our studies. First, potential variance restrictions in AAJI could limit the generalizability of our findings. Our results indicate that employees' affective appraisal of new technologies as threatening to their employment is currently confined, despite the introduction of ChatGPT between the second and third measurement points in Study 2. Nevertheless, AAJI will be of rising

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importance with ongoing digitalization. Future research is urged to investigate its (potentially nonlinear) trajectory over even longer periods.

Second, similar constructs to AAJI were recently introduced, highlighting the importance of digitalization-specific forms of job insecurity for theory and practical applications (Park et al. 2024; Roll et al. 2023). For instance, Roll et al. (2023) introduced the concept of *occupation insecurity*. Future research is urged to differentiate AAJI from these constructs empirically to extend its nomological net and to gain a better understanding of employees' affective appraisal of STARA-driven task and job substitutions. We also stress researchers to develop scales targeting cognitive automation-related job insecurity and to empirically separate it from our proposed affective counterpart to achieve a multidimensional digitalization-specific job insecurity measure.

Third, while the assessment of the substitution potential of occupations strengthens the validity of our findings, the data from 2019 might be slightly dated considering the rapid development of new technologies in recent years. We stress the replication of our findings with more current data if available and the identification of additional national/macroeconomic factors that influence AAJI (e.g., economic condition or employment protection legislation) (Shoss 2017).

Fourth, we used a self-developed single-item measure to capture the extent to which employees use STARA at work since established, validated questionnaires are missing to date. Also, future research should consider that employees can differ in their appraisal of STARA as either a resource, challenge, or hindrance demands, which also implies positive effects on work outcomes. Several researchers (e.g., Gödöllei and Beck 2023) emphasize the crucial role of appraisal of STARA as technologies can also improve performance. Instead of broadly investigating the use of STARA, we also stress the identification of potential risk technologies (e.g., embodied robots, and large language models like ChatGPT) (Yam et al. 2023).

Fifth, we neglected the investigation of specific work settings that could moderate the relationships between use of STARA and AAJI. For instance, these could include whether human and technology share work tasks in collaborative settings or do not share tasks in non-collaborative settings (Wang et al. 2023).

Sixth, we identified factors that uniquely determine AAJI and therefore emphasize the relevance of this construct in the digitalized world of work. Future research is urged to investigate its predictive validity for digitalization-specific outcomes (e.g., technology acceptance, trust in AI/automation, or technostress) or more general work-related outcomes (e.g., burnout, turnover intentions), even above cognitive and affective JI and the original STARA Awareness scale by Brougham and Haar (2018).

Finally, based on our empirical separation of AAJI from cognitive and affective JI, researchers should develop and evaluate interventions to specifically reduce AAJI instead of relying on interventions that are successful for established JI (Abildgaard et al. 2018). This seems necessary to support employees with high use of STARA in the long run and to prevent potential negative impacts of JI due to STARA with regard to health and work outcomes.

6 | Conclusion

In sum, our findings essentially contribute to research on the AAJI as an important construct to capture employees' affective appraisal of STARA-driven task and job substitution. We shed light on its construct validity, by taking a closer look at the internal structure, and its nomological net, tested potential antecedents (in terms of indicators of technological change and personality traits), and outlined its 1-year trend. Considering the rapid technological advances in all work environments, this timely construct contributes to much-needed research and practical applications.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

[1] Nomological networks describe a conceptual and empirical system that defines a construct of interest by modeling its relationships with other constructs and observable variables. They depict the theoretical and statistical relationships between a construct of interest and its measurement and relationships with other constructs and their measurements (Cronbach and Meehl 1955).

[2] Additionally, CFAs in which we separated quantitative (loss of job as a whole) and qualitative items (loss of core tasks within occupation) yielded poor model fit indices. Four-factor models, in which combinations (cognitive and quantitative, cognitive and qualitative, affective and quantitative, affective and qualitative) loaded on separate factors, also yielded poor model fit indices.

[3] Considering the adaptation of the construct, we test hypotheses 1-5 for affective automation-related job insecurity instead of STARA Awareness.

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